

PropertyScope NSW

Release 1 / Technical report

41026 Advanced Software Development — Assessment 2

Submission	Group 20
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1 Project overview and Release 1 scope

PropertyScope NSW helps buyers research NSW property. In Release 0 each student built a containerised frontend, backend/API and database feature behind a shared home page, using the host AI-mode's Plan/Act/Observe/Adapt loop. Release 1 keeps these and adds a shared MCP server, a shared RAG server and MCP/RAG loop validation modes, all host processes outside Docker. Every feature's assistant now calls its tools through MCP and answers from its own cited corpus. Tutor approval for OpenAI and PostgreSQL was reconfirmed by Matthew on 3 October.

1.1 Responsibilities

Owner	Release 1 responsibility
Group (shared; led by Matthew)	MCP and RAG servers, AI-mode adapters, loop validation modes, adoption kit, Compose host configuration and integration evidence
Matthew Shelton, F1 Property data	14 tools (two approval-gated writes), data-platform corpus, grounded assistant
Burhan Naeem, F2 Sales and market	3 market-case tools, sale-matching corpus, assistant workspace
James Huang, F3 Suburb analytics	4 suburb and crime tools, analytics corpus, map of F1 data
Michael White, F4 Due diligence	3 site-review tools, evidence-state corpus, assistant, CI tests
Derek Song, F5 Buyer workspace	5 buyer-case tools, workflow corpus, scoped assistant routes

Each student also maintains their feature's Compose services and `student-N.yml` workflow.

1.2 Delivery plan and risks

The shared runtime and Feature 1 landed first (7 September); an adoption kit (19 September) let each owner register tools, a corpus and a route independently. A final increment captured one running stack ([release audit](#)). Versioned evidence and bounded failures limited the main risks: missing data and provider availability.

2 Functional requirements

ID	Requirement	Evidence
FR1	Each feature keeps its Release 0 frontend, backend/API, database CRUD and AI-mode	6.3
FR2	MCP lists registered tools, runs only a feature's allowlisted tools and returns schema-validated results	5.1, 6.1, 6.3
FR3	RAG ingests versioned corpora and searches only the requesting feature's corpus	5.2, 6.1
FR4	Every feature frontend reaches MCP and RAG through its own backend assistant/turns route	6.3, 7
FR5	Answers show citations and a confidence category, or an insufficient-context response when nothing relevant is found	6.3
FR6	The shared loop keeps its Release 0 mode and adds separate MCP and RAG validation modes	6.2
FR7	Compose deploys the feature containers with host AI-mode, MCP and RAG addresses; no AI service is containerised	6.6
FR8	Each student-N.yml builds and tests its feature with MCP and RAG disabled	6.5

3 Non-functional requirements

A shared AI tier risks one feature reaching another's tools or passages, untraceable answers, and AI failures breaking ordinary feature work. These targets address those risks.

Quality	Requirement and target	Validation
Security	Host AI APIs need a service token; MCP and RAG bind to 127.0.0.1. Every unauthenticated call is refused.	Live probe: unauthenticated calls return 401
MCP tool boundaries	A run calls only its feature's allowlisted tools; each call is signed for feature, run, arguments and deadline. Writes need approval.	Architecture checks block cross-feature imports and database access; tests reject out-of-scope calls
RAG grounding and traceability	Every guidance finding cites a retrieved passage with document, corpus version and content hash; current facts cite a tool call.	Citation figures; corpus version recorded per run
Retrieval quality	Expected-source recall@5 ≥ 0.9 on authored F1 questions	Measured 0.963 (6.4)
Reliability	Every tool call has a deadline and a structured error; RAG failure gives insufficient context, never an unsupported answer	Outage tests with controlled doubles
Performance	At most five passages per retrieval; tool deadlines 2–10 s	Loop: MCP tool 15 ms, retrieval 108 ms
Usability	Citations and run activity are keyboard accessible and reflow in narrow panels	Browser checks
Maintainability	A feature adopts MCP and RAG by editing only its own catalogue, corpus and allowlist	All five pass catalogue and architecture checks
Interoperability	MCP Streamable HTTP with JSON Schema tool contracts	SDK round-trip tests validate schema-valid results

Quality	Requirement and target	Validation
Availability	Feature CRUD works while AI services are down; RAG history survives restarts	Outage, store-reopen and re-ingestion tests

4 Release 1 architecture

Compose runs the shared edge and five feature slices, each with a frontend, backend/API and owned database service. AI-mode (containing the agentic loop), MCP and RAG run as separate host processes and are never containerised. Backends reach AI-mode through `host.docker.internal`; AI-mode calls MCP, RAG and the OpenAI model; MCP dispatches only to registered feature routes. Neither shared server opens a feature database.

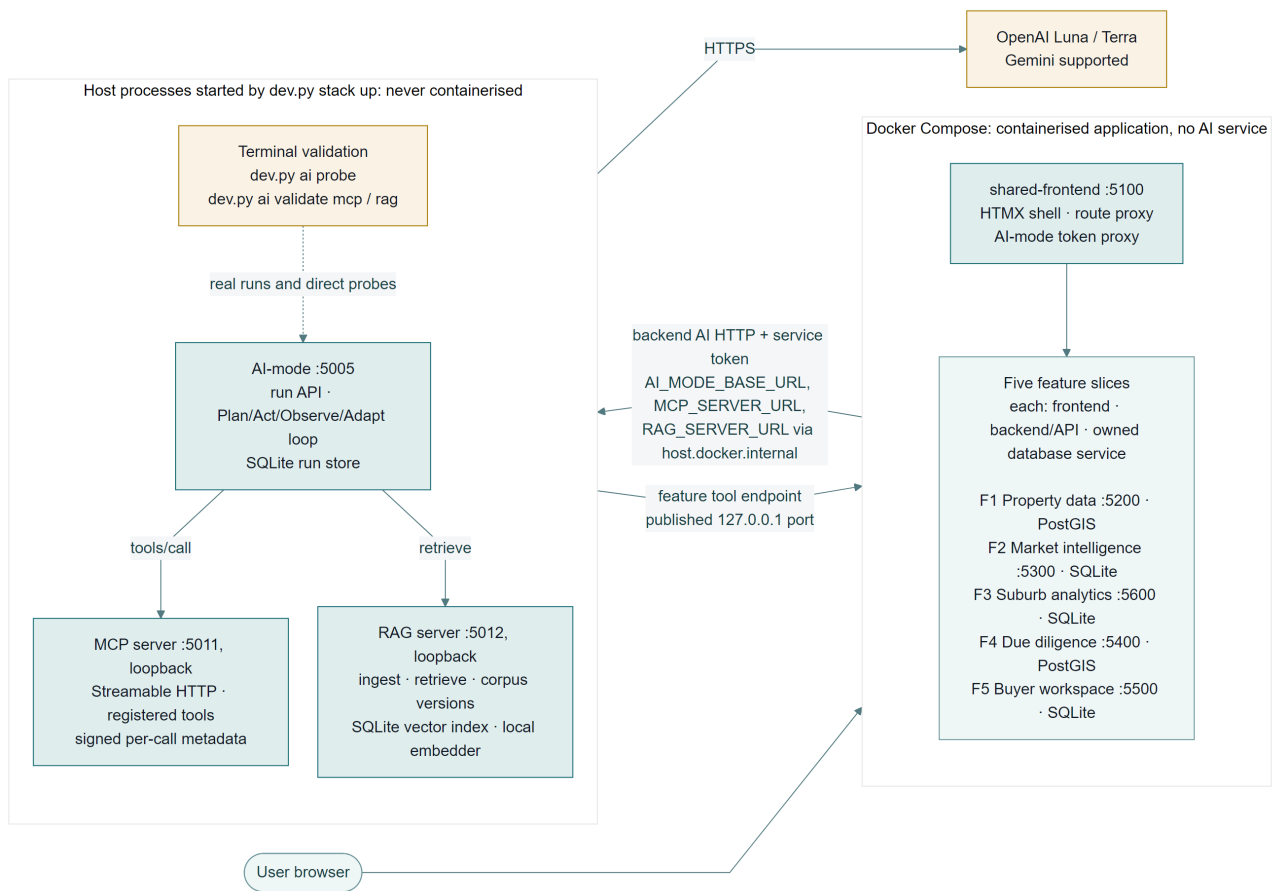


Figure 1 Overall Release 1 architecture and container boundary

4.1 Repository structure

Path	Responsibility
student-1/ ... student-5/	Owned UI, API, persistence, tools and corpus
ai-services/agent-core/	Loop policies, grounding and confidence
ai-services/ai-mode/	Run API, provider and MCP/RAG adapters
ai-services/mcp-server/, rag-server/	Tool transport; local semantic retrieval
shared/contracts/, tool-runtime/, frontend/	Schemas, dispatch and research shell
docker-compose.yml, deployment/	Release 0 Compose with host MCP/RAG addresses
.github/workflows/, scripts/	CI and developer commands

5 MCP and RAG design

A question goes from a feature's frontend to its own backend, which starts an AI-mode run fixed to that feature's tool allowlist and corpus. MCP checks each call's signed metadata, dispatches it to the owning backend and validates the result against the tool's output schema. Results carry `transport: mcp`, distinguishing them from direct HTTP.

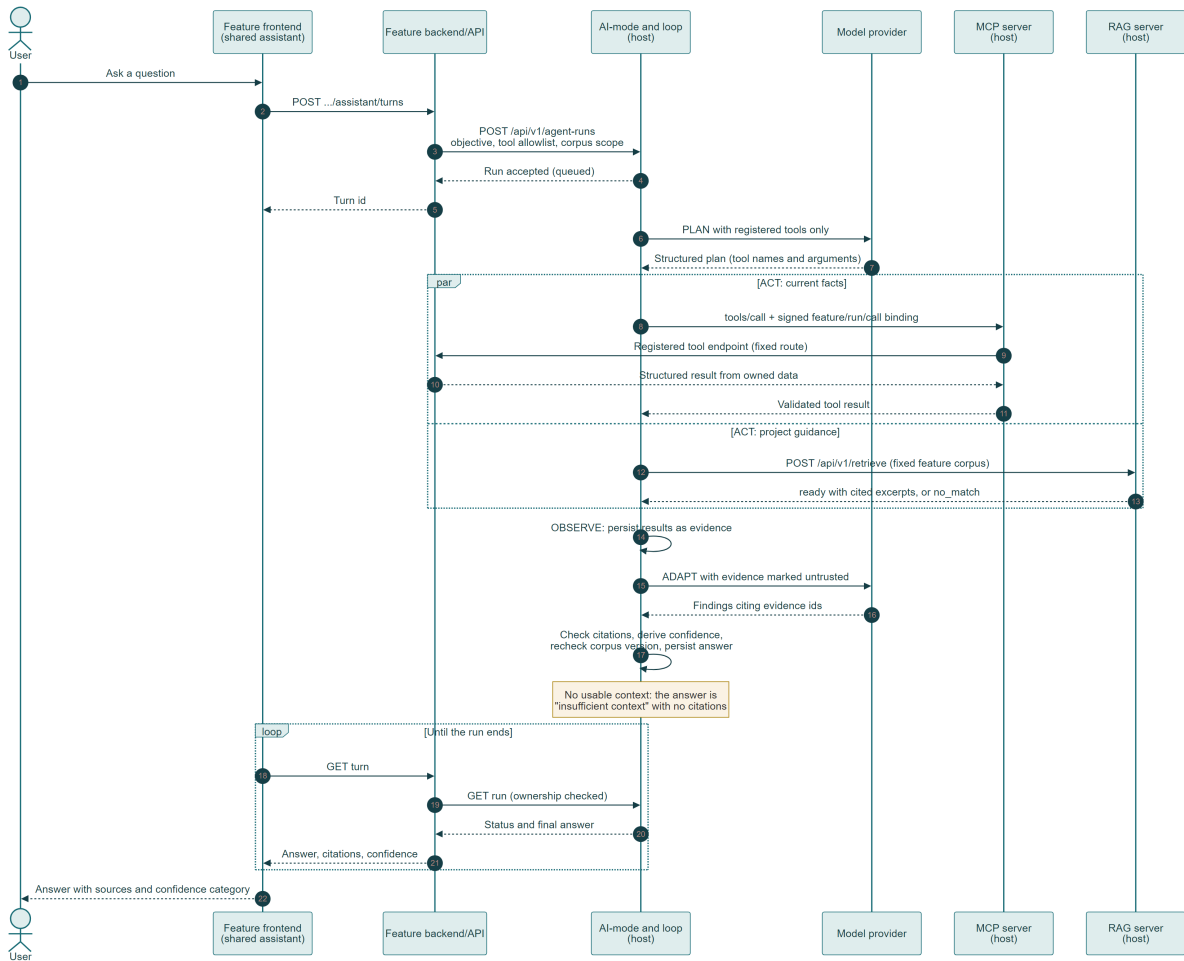


Figure 2 Frontend, backend, MCP and RAG interaction flow

5.1 Tools, inputs and outputs

MCP lists 29 feature tools; a run sees only its own feature's tools. 27 are read-only and Feature 1's two writes need human approval. `context.retrieve.v1` is AI-mode's internal retrieval step, not an MCP tool. Full JSON schemas are in each feature's `tool-catalog.yaml`.

Student	Feature	Registered MCP tools	Access	RAG corpus
1	Data Platform and Property Discovery	<code>platform.capabilities.v1</code> , <code>data.sources.v1</code> , <code>data.releases.v1</code> , <code>data.runs.v1</code> , <code>data.release_inspect.v1</code> , <code>data.run_inspect.v1</code> , <code>data.run_explain.v1</code> , <code>data.release_compare.v1</code> , <code>data.coverage.v1</code> , <code>property.search.v1</code> , <code>property.inspect.v1</code> , <code>property.locality_summary.v1</code> , <code>data.run_retry.v1</code> , <code>data.release_publish.v1</code>	12 read-only, 2 write, 2 need approval	operator-guidance (19 documents)
2	Property Sales Explorer and Market Cases	<code>market.capabilities.v1</code> , <code>market.cases.inspect.v1</code> , <code>market.sales.summary.v1</code>	3 read-only	operator-guidance (3 documents)
3	Suburb, Crime and Liveability Analytics	<code>suburb.published-context.v1</code> , <code>suburb.snapshot.v1</code> , <code>crime.compare.v1</code> , <code>crime.methodology.v1</code>	4 read-only	suburb-analytics-guidance (8 documents)
4	Site, Planning and Building Due Diligence	<code>duediligence.capabilities.v1</code> , <code>duediligence.review.inspect.v1</code> , <code>duediligence.evidence.summary.v1</code>	3 read-only	operator-guidance (3 documents)
5	Buyer Journey Workspace	<code>buyer.capabilities.v1</code> , <code>buyer.cases.inspect.v1</code> , <code>buyer.notes.list.v1</code> , <code>buyer.tasks.list.v1</code> , <code>buyer.evidence.collect.v1</code>	5 read-only	operator-guidance (5 documents)
Shared	Grounded runs	<code>context.retrieve.v1</code> , added by AI-mode to runs of features with a registered corpus	Read-only	The feature's own corpus

Tool	Required inputs	Output fields	Side effect	Timeout
<code>platform.capabilities.v1</code>	None	<code>revision</code> , <code>application</code> , <code>assistant</code> , <code>features</code> , <code>suggested_questions</code> , <code>feature_status_meaning</code>	read-only	3000 ms
<code>data.sources.v1</code>	None	<code>items</code> , <code>count</code>	read-only	3000 ms
<code>data.releases.v1</code>	None	<code>items</code> , <code>count</code>	read-only	3000 ms
<code>data.runs.v1</code>	None	<code>items</code> , <code>count</code>	read-only	3000 ms
<code>data.release_inspect.v1</code>	<code>release_id</code>	<code>release</code> , <code>quality_results</code> , <code>quality_summary</code> , <code>receipts</code> , <code>activations</code> , <code>consumer_imports</code> , <code>publication_policy</code> , <code>accepted_predecessor</code> , <code>release_contract</code>	read-only	3000 ms
<code>data.run_inspect.v1</code>	<code>run_id</code>	<code>run</code> , <code>tasks</code> , <code>quality_results</code>	read-only	3000 ms

Tool	Required inputs	Output fields	Side effect	Timeout
data.run_explain.v1	run_id	run, tasks, quality_results, insight	read-only	3000 ms
data.release_compare.v1	candidate_release_id, predecessor_release_id	candidate, predecessor, differences	read-only	3000 ms
data.coverage.v1	locality	items	read-only	3000 ms
property.search.v1	query	items, count	read-only	3000 ms
property.inspect.v1	property_ref	property, identifiers, aliases, coverage, sales_history	read-only	3000 ms
property.locality_summary.v1	None	availability, scope, total_registered_addresses, unit_number_summary, bounding_box, top_streets, release	read-only	5000 ms
data.run_retry.v1	run_id, idempotency_key	child_run_id, replayed	reversible-write, approval	5000 ms
data.release_publish.v1	release_id, idempotency_key	status, receipt_id, replayed	destructive-write, approval	5000 ms
market.capabilities.v1	None	revision, feature, tools, limitations, suggested_questions	read-only	3000 ms
market.cases.inspect.v1	market_case_id	market_case	read-only	3000 ms
market.sales.summary.v1	market_case_id	summary, sales	read-only	3000 ms
suburb.published-context.v1	locality	locality, status, population, schools, crime_coverage, sources, limitations	read-only	4000 ms
suburb.snapshot.v1	locality	suburb, places, limitations	read-only	3000 ms
crime.compare.v1	localities, from_month, to_month, measure	series, measure, offence, source_release, limitations	read-only	4000 ms
crime.methodology.v1	None	geography, measures, offence_categories, zero_missing_rule, rate_method, responsible_use, source_release	read-only	2000 ms
duediligence.capabilities.v1	None	revision, feature, tools, limitations, suggested_questions	read-only	3000 ms
duediligence.review.inspect.v1	site_review_id	site_review	read-only	3000 ms
duediligence.evidence.summary.v1	site_review_id	constraints, buildings	read-only	3000 ms
buyer.capabilities.v1	None	feature_key, purpose, limitations, suggested_questions, read_only	read-only	5000 ms

Tool	Required inputs	Output fields	Side effect	Timeout
buyer.cases.inspect.v1	buyer_case_id	buyer_case, shortlisted_properties, property_count, limit, content_is_untrusted	read-only	5000 ms
buyer.notes.list.v1	buyer_case_id	items, count, limit, content_is_untrusted	read-only	5000 ms
buyer.tasks.list.v1	buyer_case_id	items, count, limit, content_is_untrusted	read-only	5000 ms
buyer.evidence.collect.v1	buyer_case_id	state, sections, evidence_references, limitations, bounds, content_is_untrusted	read-only	10000 ms

5.2 Knowledge sources and grounded responses

Each feature has its own corpus of authored CC0 project guidance; current facts come from MCP tools. Manifests list every source: [F1](#), [F2](#), [F3](#), [F4](#), [F5](#). F1 covers publication and data limits (19 documents); F2 sale matching and case eligibility (3); F3 zero versus missing evidence (8); F4 site-review evidence states (3); F5 case stages, shortlists and tasks (5).

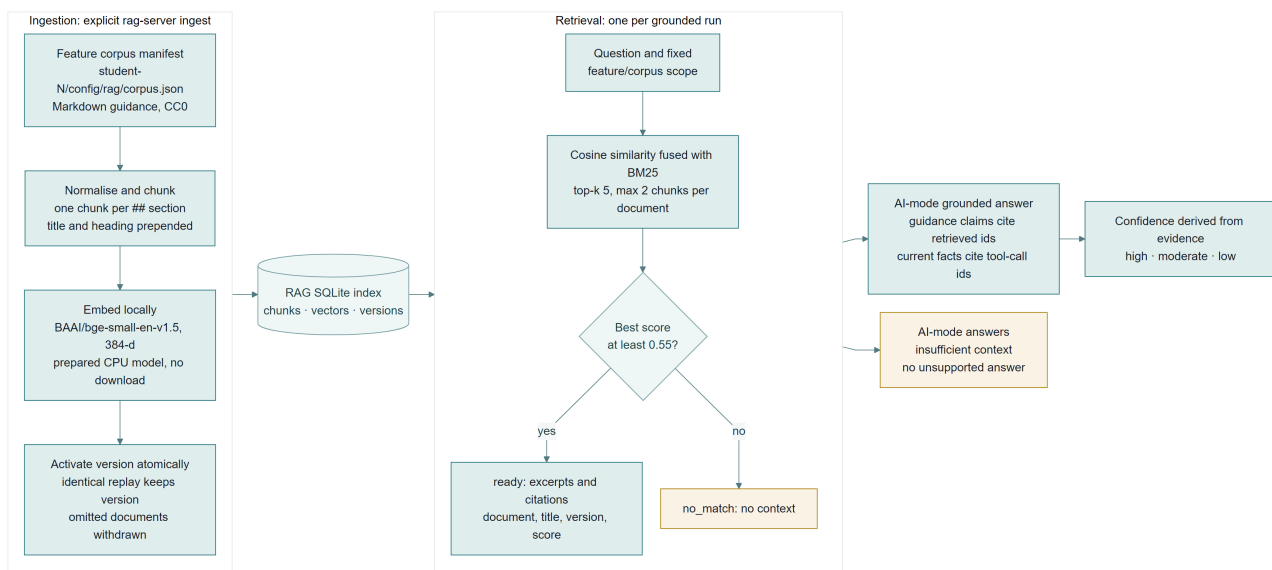


Figure 3 Versioned ingestion, retrieval and grounding

Passages are embedded locally with BGE-small. Retrieval fuses semantic and keyword ranking, returns at most five passages (two per document) and drops anything below a 0.55 relevance score. Retrieved text is treated as untrusted evidence. Confidence is derived from the evidence, not reported by the model: **high** needs two or more supports (documents or tool results), strong matches and no gaps; gaps, weaker matches or a single support give **moderate**; **low** marks stale or conflicting evidence; **insufficient** means no relevant context and carries no citations.

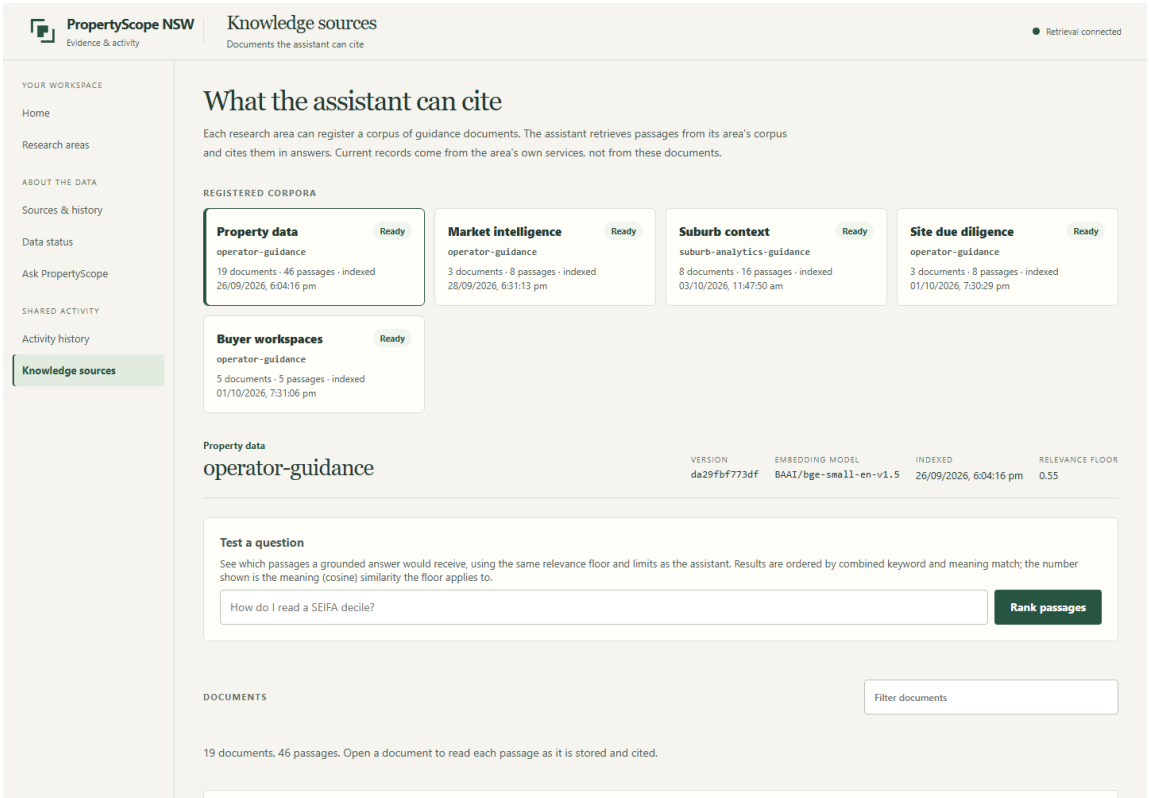


Figure 4 Registered knowledge sources in the running application

6 Validation and results

6.1 Terminal validation

On 3 October the [live probe](#) checked the running host services directly from a terminal:

```
$ uv run scripts/dev.py ai probe
MCP: HTTP 200; 29 tools; feature counts 14 / 3 / 4 / 3 / 5
RAG: BAAI/bge-small-en-v1.5 ready; 384 dimensions
Corpora: 19 / 3 / 8 / 3 / 5 documents
Each feature: relevant retrieval PASS; off-topic no_match PASS
Unauthenticated AI-mode / MCP / RAG: HTTP 401
All checks passed.
```

6.2 Shared agentic loop

ai validate mcp and ai validate rag run the production Plan/Act/Observe/Adapt loop against the live MCP and RAG servers. Model decisions are scripted so each run is repeatable; Section 6.3 shows real model answers. The Release 0 mode remains available.

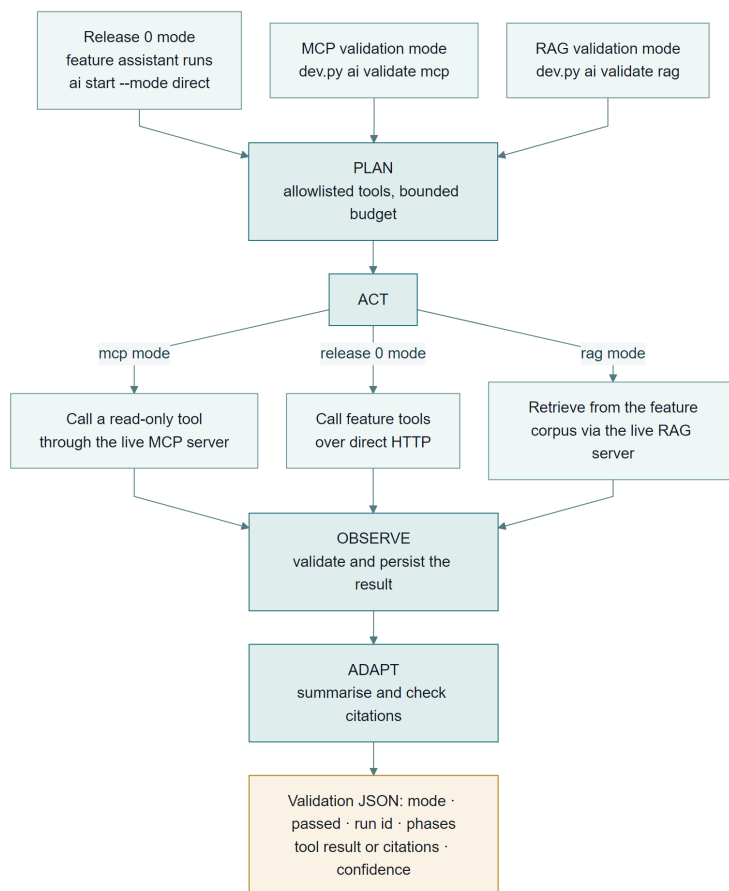


Figure 5 Loop entry points and validation modes

```

$ uv run scripts/dev.py ai validate mcp
mode      mcp  passed=true  status=succeeded
feature   student-1-propertyscope-data-platform
tool      platform.capabilities.v1
corpus    -
run_id    97ea140e-2a1c-4f23-a1bf-c1a4f6715432
request_id r1-mcp-654a814d-bab5-4b34-9d09-7eac0adf450d
phases    plan -> act -> observe -> adapt
transport live-local-services  decisions=deterministic-validation
tool call  f5b53440-94a7-4730-b3a2-d8b397e969e0: succeeded in 15 ms; fields: application,
assistant, feature_status_meaning, features
summary    Registered MCP tool returned structured evidence.
boundary   Production runner and local services; deterministic model decisions, not provider
answer quality.
  
```

```

$ uv run scripts/dev.py ai validate rag
mode      rag  passed=true  status=succeeded
feature   student-1-propertyscope-data-platform
tool      context.retrieve.v1
corpus    operator-guidance
run_id    3e319a2e-1667-4f36-9397-2f7502d8fd34
request_id r1-rag-073dd32b-de85-44ec-a822-ba01bb09a474
phases    plan -> act -> observe -> adapt
transport live-local-services  decisions=deterministic-validation
tool call  0516ef7b-bbf9-474b-a646-2ebbb9522c63: succeeded in 108 ms, retrieval ready, 5 citations
version    da29fbf773df88ce8a0939cae9f8ade1ca094441e368ac3fba2ac6c9c1cb1fed
cites      publication: Publishing a version
  
```

cites glossary: Property data terms
cites data-updates: Running a data update
final confidence=moderate grounding=ready
summary Retrieved project guidance.
boundary Production runner and local services; deterministic model decisions, not provider
answer quality.

6.3 Feature UI, API and persistence

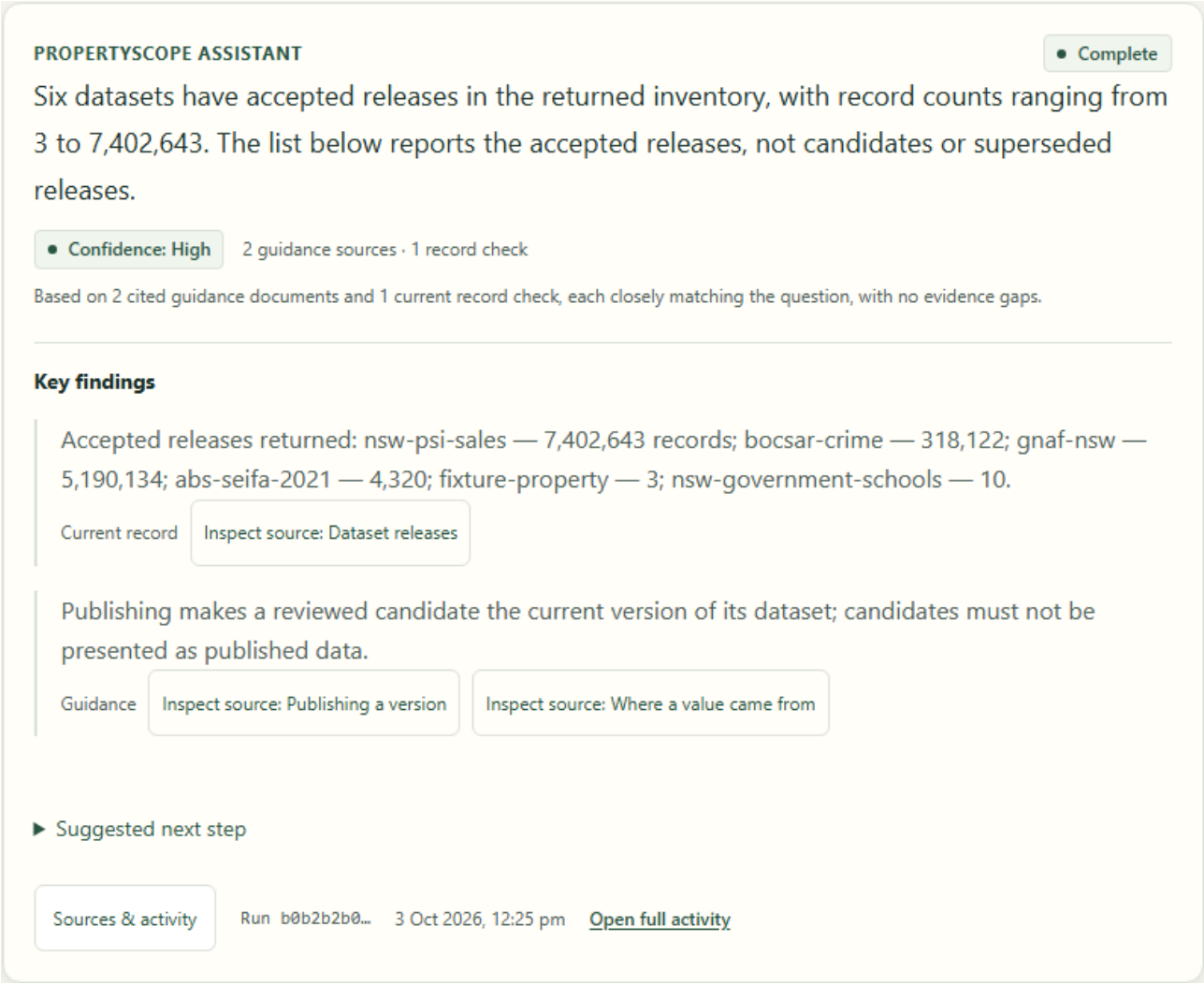
The [live operations record](#) checked every feature on the running stack. F1 and F4 created, changed and deleted a labelled temporary record; the others report rows in their owning store, including demonstration records.

Feature page	Backend API read	Owned database operation
1: HTTP 200	/api/data-platform/v1/sources 200, 10 items	Create, read, update, delete (201/200/200/200/204); read after delete 404
2: HTTP 200	/api/market-intelligence/v1/market-cases 200, 10 items	Rows: market_case 10, sale_observation 26
3: HTTP 200	/api/suburb-analytics/v1/suburbs 200, 10 items	Rows: suburb_info 10, suburb_indicators 400, suburb_amenity 38, suburb_overview 10, user_suburbs 10
4: HTTP 200	/api/due-diligence/v1/site-reviews 200, 10 items	Create, read, update, delete (201/200/200/200/200); read after delete 404
5: HTTP 200	/api/buyer-workspaces/v1/buyer-cases 200, 10 items	Rows: buyer_case 10, case_note 11, case_property 12, case_task 12

Each assistant then answered three questions through its own UI with the live OpenAI provider: current records via MCP, cited guidance via RAG, and an off-topic question. Captions summarise each question; full captures show its wording, and run metadata records tool transports.

Feature 1 Property data

Full UI captures: [mcp](#), [rag](#), [insufficient](#). [Run metadata](#).



PROPERTYSCOPE ASSISTANT

● Complete

A property can show no recorded sales because PropertyScope only displays published sale records that match exactly one G-NAF address. An empty sale history is missing matched evidence, not proof that the property has never sold.

● Confidence: Moderate

2 guidance sources

Based on 2 cited guidance documents; the answer notes 1 evidence gap.

Key findings

Sales with ambiguous or unrecognised addresses are retained as unmatched rather than linked by guesswork. Older free-text records and unit sales are more likely to remain unmatched; a sale may also be older than 1990 or not yet published.

Guidance

Inspect source: NSW property sales (PSI)

“No published sale history” means no matched sale was found in the published NSW sales data; it does not establish that no sale occurred.

Guidance

Inspect source: Reading a property record

Evidence gaps

No exact property reference was supplied, so the cause for a particular property's missing sale history cannot be established.

► Suggested next step

Sources & activity

Run 421e03a2... 3 Oct 2026, 12:26 pm [Open full activity](#)

Figure 7 F1 RAG: why a sold property can lack sales

▼ NSW property sales (PSI)

● Project guidance

Project guidance explains the workflow; it does not establish current property conditions.

How sales are matched to a property

A sale is linked to a property record only when its address resolves to exactly one published G-NAF address. The match is shown with its method and confidence on the Sale history tab. If the address is ambiguous or not found, the sale is kept but left unmatched rather than guessed.

Matching is incomplete. In the September 2026 load, about 69% of post-2001 sale records and 54% of pre-2001 records matched a single address. Unit sales and older records written in free-text address formats are the most likely to stay unmatched.

An empty Sale history tab means no matched sale was found in the published sales data. It does not mean the property has never sold: the sale may be unmatched, older than 1990, or not yet published.

Location
Property data guidance: sales-history; section How sales are matched to a property; chars 785-1554

Source date
2026-09-26

Indexed
26 Sept 2026, 6:04 pm

Corpus
operator-guidance

Version
da29fbf773df88ce8a0939cae9f8ade1ca094441e368ac3fba2ac6c9c1cb1fed

Document / chunk
sales-history / 314ac51b545f054de3370a284397d237513cd50947e6323c3e0f1a0217c80018

Citation ID
cite-f4c9c4313f6a963f1897eace96cc3f316c1bcaa0a501cf9194e4052c416391b7

Content hash
a2a23bd56b60ca531cd994565d677166838979584ec34c474d74106bff31d7ef

[Open source document \(new tab\)](#)

Figure 8 F1 cited passage

PROPERTYSCOPE ASSISTANT

Complete

I can't advise on oven temperature for a chocolate cake because this assistant only supports PropertyScope property-data research. Check the specific cake recipe's baking instructions.

Confidence: Insufficient

1 record check

Relevant document context is unavailable or insufficient.

Key findings

This assistant can inspect only Property data and its available capabilities concern NSW property research, rather than cooking guidance.

Current record

Inspect source: Available research tools

Evidence gaps

Similarity measures retrieval relevance, not factual confidence

No grounded cooking or recipe guidance was retrieved for chocolate-cake oven temperatures.

Suggested next step

Sources & activity

Run 78ca25c8... 3 Oct 2026, 12:26 pm [Open full activity](#)

Figure 9 F1 off-topic: cake oven temperature

Feature 2 Market cases

Full UI captures: [mcp](#), [rag](#), [insufficient](#). [Run metadata](#).

PROPERTYSCOPE ASSISTANT

Complete

Sydney example – full history for 11 Example Street, Sydney NSW 2000 has 0 eligible sales. No median recorded price is available because there are no eligible sale records.

Confidence: High

2 record checks

Based on 2 current record checks, each closely matching the question, with no evidence gaps.

Key findings

The case has 0 eligible sales, 0 excluded sales, no transaction-volume entries and no source releases for the 2019–2026 period at the selected tier B threshold. The median recorded price is unavailable.

Current record

Inspect source: Market sales summary

The inspected case is Sydney example – full history for 11 Example Street, Sydney NSW 2000.

Current record

Inspect source: Market cases inspect

With fewer than three eligible sales, the evidence is insufficient; recorded sales are historical evidence and not a valuation or buying recommendation.

Current record

Inspect source: Market sales summary

Suggested next step

Sources & activity

Run a31a499d... 3 Oct 2026, 12:26 pm [Open full activity](#)

Figure 10 F2 MCP: case eligible sales and median price

PROPERTYSCOPE ASSISTANT

Complete

The minimum match tier excludes sales whose address-match tier is below the selected threshold.

Sydney example – full history for 11 Example Street, Sydney NSW 2000 uses tier B and currently has no eligible recorded sales, so its summary cannot establish a representative market figure.

Confidence: Moderate

2 guidance sources · 2 record checks

Based on 2 cited guidance documents and 2 current record checks; the answer notes 1 evidence gap.

Key findings

A minimum tier of B excludes sales matched below tier B. Tier A is stricter; increasing the threshold narrows the sample to more confidently attributed address matches, while lower tiers may include neighbouring or similarly named addresses. The tier measures address-match confidence, not whether a price is representative.

Guidance

Inspect source: How sales are attributed to a property and how confident that is

The case covers 1 January 2019 to 31 December 2026 at tier B. It has 0 eligible sales, 0 recorded exclusions, no median recorded price, no transaction-volume entries and 0 source releases.

Current record

Inspect source: Market cases inspect

Inspect source: Market sales summary

Few eligible sales can result from a strict match threshold, or records excluded because they lack a contract date, fall outside the case period, have a missing or zero price, or fall below the threshold. It does not by itself show a quiet market.

Guidance

Inspect source: What Sales and market evidence cannot establish

The summary labels fewer than three eligible sales as insufficient evidence and states that recorded sales are historical evidence, not a valuation or buying recommendation.

Current record

Inspect source: Market sales summary

Evidence gaps

There are no source releases or sale records in this summary, so it cannot show why no records were available beyond the reported zero exclusions.

Suggested next step

Sources & activity

Run c6f47a81... 3 Oct 2026, 12:26 pm [Open full activity](#)

Figure 11 F2 RAG: what the match tier excludes

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▼ What Sales and market evidence cannot establish

● Project guidance

Project guidance explains the workflow; it does not establish current property conditions.

Coverage gaps and excluded records

A summary reports only the sales that passed the case filters, and separately counts what it left out. Records are excluded for four reasons: no contract date, a contract date outside the case period, a missing or zero price, or a match tier below the threshold you selected.

Excluded records are not errors and are not deleted. Raising the minimum match tier deliberately narrows the evidence, so a case with few eligible sales may reflect a strict threshold rather than a quiet market.

Where fewer than three sales are eligible, the summary states that this is insufficient evidence. Treat the median from such a case as an observation about those specific records, not a representative figure.

Location
Operator guidance: evidence-limits; section Coverage gaps and excluded records; chars 1169-1905

Source date
2026-09-28

Indexed
28 Sept 2026, 6:31 pm

Corpus
operator-guidance

Version
fc79eb7f58187a25419ecc32b714dfb061d7e9c4e3233358bad4c98101b881d2

Document / chunk
evidence-limits / ecd74b4739d4def521295212538ff53ccf38197be842e39ec94e3622e67cd369

Citation ID
cite-c70c0ee60171876d19c9f70e65ddc7be023393abc58337a9b19395058faac697

Content hash
22a036de43173fa7388afdd5b2072a4844c459d69209a081d7439d7be7e78d29

[Open source document \(new tab\)](#)

Figure 12 F2 cited passage

PROPERTYSCOPE ASSISTANT

• Complete

Insufficient context to provide a grounded explanation.

• Confidence: Insufficient

Relevant document context is unavailable or insufficient.

Evidence gaps

Similarity measures retrieval relevance, not factual confidence

Password-reset instructions or an authorised account-support capability were not available.

► Suggested next step

Sources & activity

Run 0dbebe76... 3 Oct 2026, 12:26 pm [Open full activity](#)

Figure 13 F2 off-topic: password reset

Feature 3 Suburb analytics

Full UI captures: [mcp](#), [rag](#), [insufficient](#). [Run metadata](#).

PROPERTYSCOPE ASSISTANT

● Complete

A recorded zero is 0 only within the source’s declared observation universe; missing or unavailable evidence is null and must not be treated or displayed as zero. This feature provides count and rate measures; its fixture rate is count divided by fixture population, multiplied by 100,000.

● Confidence: Moderate

2 guidance sources · 1 record check

Based on 2 cited guidance documents and 1 current record check; the answer notes 1 evidence gap.

Key findings

The feature’s methodology defines a recorded zero as 0 and missing evidence as null. Missing or unavailable evidence must not be plotted or described as zero, and sparse coverage does not justify assumptions about unobserved months.

Guidance

Inspect source: How to interpret recorded crime evidence

Inspect source: Published evidence and demonstration data

The available measures are count and rate. The configured fixture calculation is: fixture count ÷ fixture population × 100,000; the source release is demo-2026.1.

Current record

Inspect source: Crime methodology

A count is recorded observations for the selected categories and period, while a rate uses a stated population denominator. They are not interchangeable, and comparisons must not mix them; differing count and population reference periods should be disclosed.

Guidance

Inspect source: How to interpret recorded crime evidence

The methodology response describes an NSW suburb fixture, so its rate method and release identify demonstration data rather than current NSW conditions.

Guidance

Inspect source: Published evidence and demonstration data

Evidence gaps

No published locality-specific population denominator, observation universe or reference period was provided for calculating a current locality rate.

► Suggested next step

Sources & activity

Run 820c8083... 3 Oct 2026, 12:27 pm [Open full activity](#)

Figure 14 F3 MCP: crime methodology and rate rules

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PROPERTYSCOPE ASSISTANT

● Complete

Feature 3 displays a recorded zero as 0 only where the source observed the relevant category and period. Missing or unavailable evidence remains null, not zero.

● Confidence: High2 guidance sources · 1 record check

Based on 2 cited guidance documents and 1 current record check, each closely matching the question, with no evidence gaps.

Key findings

The Feature 3 methodology rule is: recorded zero is 0; missing evidence is null. The methodology source release is demo-2026.1 and applies to an NSW suburb fixture.

Current recordInspect source: Crime methodology

A zero is meaningful only within the source’s declared observation universe. Missing or unavailable evidence must not be plotted or described as zero, and sparse coverage does not justify assumptions about unobserved months.

GuidanceInspect source: How to interpret recorded crime evidence

Unavailable evidence is not evidence that an event or other characteristic does not exist.

GuidanceInspect source: Published evidence and demonstration data

► Suggested next step

Sources & activity

Run cc21e6f6...3 Oct 2026, 12:27 pmOpen full activity

Figure 15 F3 RAG: recorded zero versus missing

▼ How to interpret recorded crime evidence

• Project guidance

Project guidance explains the workflow; it does not establish current property conditions.

Zero and missing

A recorded zero is meaningful only inside the source's declared observation universe. Missing or unavailable evidence remains null and must not be plotted or described as zero. Sparse coverage does not support an assumption about unobserved months.

Responsible interpretation

Recorded crime data does not establish why events occurred, an individual's risk, the safety of a place or future crime. A comparison is descriptive evidence for a controlled period and category, not a suburb ranking.

Location

Suburb analytics guidance: crime methodology; section Zero and missing; chars 412-930

Source date

2026-10-02

Indexed

3 Oct 2026, 11:47 am

Corpus

suburb-analytics-guidance

Version

8db8f59a9c9daa8ffd5126f157fa0097a1a83bf9208a2a6c9d81f41321ba3b4a

Document / chunk

crime-methodology / 454002b8ce681cb06eab0023edd13ac757dc8781f0763788a22d448a2bb5c24d

Citation ID

cite-1ba53d6262ea64ce9b5a200b9d02bedec4231df4e225bc7e8da2cf6b21e65a47

Content hash

f6c34c70ee3990754b221ff9fd1f6aba2f3d116d608e995846ed31924409efde

[Open source document \(new tab\)](#)

Figure 16 F3 cited passage

PROPERTYSCOPE ASSISTANT

Complete

Insufficient context to provide a grounded explanation.

Confidence: Insufficient

Relevant document context is unavailable or insufficient.

Evidence gaps

Similarity measures retrieval relevance, not factual confidence

No relevant cooking-temperature guidance is available in the retrieved corpus.

Suggested next step

Sources & activity

Run 75389ab8... 3 Oct 2026, 12:28 pm [Open full activity](#)

Figure 17 F3 off-topic: cake oven temperature

Feature 4 Due diligence

Full UI captures: [mcp](#), [rag](#), [insufficient](#). [Run metadata](#).

PROPERTYSCOPE ASSISTANT

Complete

For 11 Example Street, Sydney NSW 2000, the review records planning controls, environmental map checks, strata information and building-related records in several evidence states. Some records are confirmed, while others have partial coverage, no mapped intersection or are unavailable.

Confidence: Moderate

2 guidance sources · 1 record check

Based on 2 cited guidance documents and 1 current record check; the answer notes 1 evidence gap.

Key findings

Planning evidence records confirmed R2 Low Density Residential zoning and a maximum floor-space ratio of 0.5:1. A maximum building height of 9.5 m has partial coverage, and a separate zoning map record is marked non-intersection.

Current record

Inspect source: Duediligence evidence summary

Environmental evidence records no intersection with the 1% AEP flood planning area. Bushfire-prone-land mapping (Vegetation Category 1) and the heritage listing record are unavailable, so the review cannot conclude either matter.

Current record

Inspect source: Duediligence evidence summary

Strata and building evidence includes a confirmed registered strata scheme with an active managing agent and a confirmed building-order or rectification-notice record. A further strata record is unavailable; tribunal matters are marked non-intersection; and written undertakings have partial coverage.

Current record

Inspect source: Duediligence evidence summary

Confirmed means a source record applies to the property; partial coverage means only part of the expected matter is covered; non-intersection means a mapped layer did not intersect the property; and unavailable means no conclusion can be drawn either way.

Guidance

Inspect source: How due-diligence evidence is classified and how confident that is

This review records gathered evidence only. It does not certify planning or Building Code compliance, safety, structural condition, approval history or legal suitability.

Guidance

Inspect source: What Site and due diligence evidence cannot establish

Evidence gaps

The unavailable bushfire, heritage and strata records do not establish the underlying position for those matters.

► Suggested next step

Sources & activity

Run 9fce0ba9... 3 Oct 2026, 12:35 pm [Open full activity](#)

Figure 18 F4 MCP: the review's recorded evidence

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PROPERTYSCOPE ASSISTANT

Complete

In a site review, confirmed means a source has an applicable record; non-intersection means a checked mapped layer did not overlap the property; partial means only part of the expected coverage is present; and unavailable means no conclusion can be drawn from that source. For 11 Example Street, each state appears in the planning, environmental, strata or building evidence, but none establishes legal suitability, compliance or safety.

Confidence: High

2 guidance sources · 1 record check

Based on 2 cited guidance documents and 1 current record check, each closely matching the question, with no evidence gaps.

Key findings

Confirmed means the source holds a record that applies to the property. At 11 Example Street, the review has confirmed zoning and floor-space-ratio controls, plus confirmed strata and building-order records.

Guidance

Inspect source: How due-diligence evidence is classified and how confident that is

Non-intersection means a mapped constraint layer was checked and did not intersect the property. The review records this state for the flood planning area and a zoning-control entry; it reduces concern for that specific layer only.

Guidance

Inspect source: How due-diligence evidence is classified and how confident that is

Inspect source: How due-diligence evidence is classified and how confident that is

Partial coverage means the source applies only partly to the property or addresses some, but not all, expected matters. The review marks building-height controls and written undertakings as partial coverage.

Guidance

Inspect source: How due-diligence evidence is classified and how confident that is

Unavailable means the source could not be reached or holds no record, so it is not evidence that a constraint is absent. The review has unavailable bushfire, heritage and strata-status evidence.

Guidance

Inspect source: How due-diligence evidence is classified and how confident that is

Inspect source: What Site and due diligence evidence cannot establish

The review evidence contains confirmed, non-intersection, partial-coverage and unavailable entries across planning, environmental, strata and building matters for 11 Example Street.

Current record

Inspect source: Duediligence evidence summary

Suggested next step

Sources & activity

Run 68f55947... 3 Oct 2026, 12:35 pm [Open full activity](#)

Figure 19 F4 RAG: evidence-state meanings

▼ How due-diligence evidence is classified and how confident that is

● Project guidance

Project guidance explains the workflow; it does not establish current property conditions.

How due-diligence evidence is classified

Every planning, environmental, strata and building record in a review carries an evidence state and a confidence, so a reader can see not just what was found but how firmly it is established. The states used are confirmed, partial coverage, non-intersection and unavailable.

What each evidence state means

Confirmed means the source holds a record that applies to this property, such as a control that covers the land or a building record on file. Partial coverage means the source applies to the property only in part, or covers some of the matters you would expect but not all of them.

Non-intersection means a mapped constraint layer, such as a flood or bushfire category, was checked against the property and did not intersect it. Unavailable means the source could not be reached or holds no record, so nothing can be concluded either way. Confidence describes how sure the source is about its own record; it is not a statement of legal or structural certainty.

Location

Operator guidance: evidence-states; section How due-diligence evidence is classified; chars 1-1016

Source date

2026-10-01

Indexed

1 Oct 2026, 7:30 pm

Corpus

operator-guidance

Version

b50761eebb33c163daecab4e0ab7fc4dbf662c69a64728d509f740479036c60e

Document / chunk

evidence-states / c4e8665526d09b2dc445bd1f879670eddd2329a65c1c5b8df06600ce6866770c

Citation ID

cite-670b34f073e817dd67b87c675ae7240f35cc43ff60e7f6c2393de999a5882e5b

Content hash

78e1d7f2add1676ca291128d18a1f0e01900dc77f3fa6738f850d3babbed0851

[Open source document \(new tab\)](#)

Figure 20 F4 cited passage

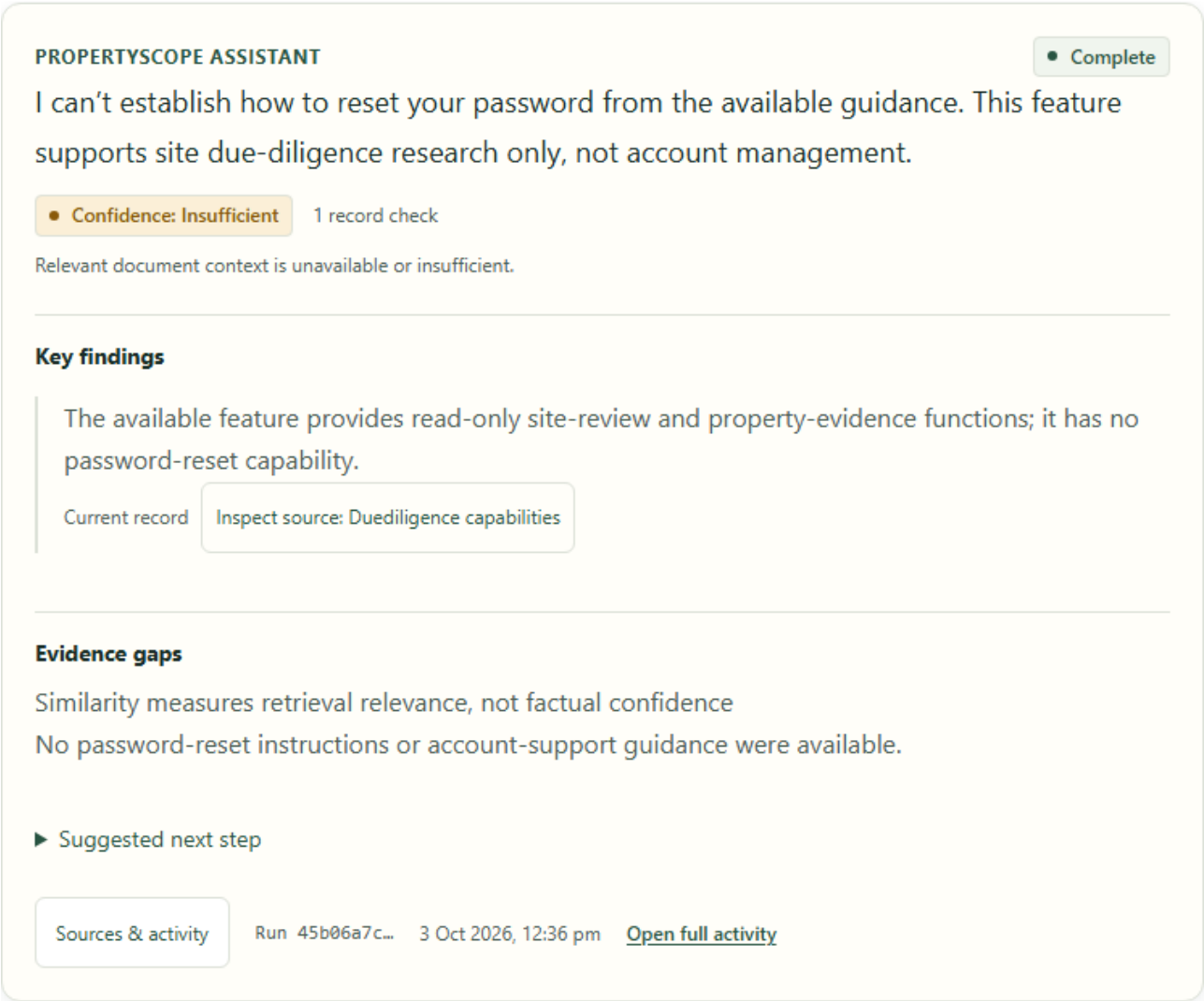


Figure 21 F4 off-topic: password reset

Feature 5 Buyer workspace

Full UI captures: [mcp](#), [rag](#), [insufficient](#). [Run metadata](#).

PROPERTYSCOPE ASSISTANT

Complete

The recorded budget is \$850,000–\$1,250,000, with SYDNEY, NSW as the target suburb. The shortlist contains 11 Example Street, Sydney NSW 2000 and 2 Sample Avenue, Mosman NSW 2088.

Confidence: Moderate

1 guidance source · 2 record checks

Based on 1 cited guidance document and 2 current record checks; the answer notes 1 evidence gap.

Key findings

The active case records an apartment preference, priorities of transport and planning evidence, a budget range of \$850,000 to \$1,250,000, and the target suburb SYDNEY, NSW.

Current record

Inspect source: Buyer cases inspect

The validated shortlist contains 11 Example Street, Sydney NSW 2000 (Reviewing; high priority; rating 4) and 2 Sample Avenue, Mosman NSW 2088 (Shortlisted; medium priority; rating 3).

Current record

Inspect source: Buyer cases inspect

Two tasks remain incomplete: confirm unassessed priorities by 15 September 2026, and verify due-diligence coverage with a qualified professional for 11 Example Street by 12 September 2026.

Current record

Inspect source: Buyer tasks list

Shortlist stages organise the buyer workflow, and ratings and priorities are personal assessments rather than valuations.

Guidance

Inspect source: Shortlists and journey stages

Evidence gaps

Property discovery evidence needs verification, Sales research evidence is conflicting, Suburb analytics evidence is unavailable, and Due diligence evidence is partial. The target suburb is a buyer preference, not proof of a property location or suitability.

Suggested next step

Sources & activity

Run 294a29c7... 3 Oct 2026, 12:29 pm [Open full activity](#)

Figure 22 F5 MCP: case budget, suburbs and shortlist

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PROPERTYSCOPE ASSISTANT

Complete

Journey stages organise your shortlist workflow: Shortlisted, Inspecting, Reviewing, Offer Considered and Closed. Your active Sydney apartment case has two validated properties: one Shortlisted and one Reviewing; use the stages, priorities and ratings to focus review work, not as value or suitability judgments.

Confidence: Moderate

2 guidance sources · 3 record checks

Based on 2 cited guidance documents and 3 current record checks; the answer notes 2 evidence gaps.

Key findings

Guidance

Inspect source: Shortlists and journey stages

The workspace stages are Shortlisted, Inspecting, Reviewing, Offer Considered and Closed. They organise workflow only: changing a stage does not arrange an inspection, submit an offer or buy a property. Ratings and priorities are personal assessments, not valuations.

Current record

Inspect source: Buyer cases inspect

The active case targets Sydney, NSW, seeks an apartment, and records transport and planning evidence as priorities. Its budget preference is AUD 850,000–1,250,000.

Current record

Inspect source: Buyer cases inspect

The shortlist contains two validated properties: 2 Sample Avenue, Mosman is Shortlisted with medium priority and a rating of 3; 11 Example Street, Sydney is Reviewing with high priority and a rating of 4.

Current record

Inspect source: Buyer tasks list

Two tasks remain incomplete: confirm unassessed priorities, and verify due-diligence coverage with a qualified professional for the Reviewing property. One sales-exclusions review task is complete.

Current record

Inspect source: Buyer evidence collect

Current cross-feature evidence is partial: Property discovery needs verification, Sales research is conflicting, Suburb analytics is unavailable, and Due diligence is partial.

Guidance

Inspect source: Buyer notes and tasks

Incomplete tasks remain visible work to review, and completing a task changes only the saved buyer record rather than carrying out external work.

Evidence gaps

Property discovery requires verification; Sales research contains conflicting information; Suburb analytics is unavailable; and Due diligence coverage is partial.

The collected evidence is bounded and does not establish valuation, affordability, condition or suitability.

Suggested next step

Sources & activity

Run 3a3e52a1... 3 Oct 2026, 12:29 pm [Open full activity](#)

Figure 23 F5 RAG: journey stages and shortlists

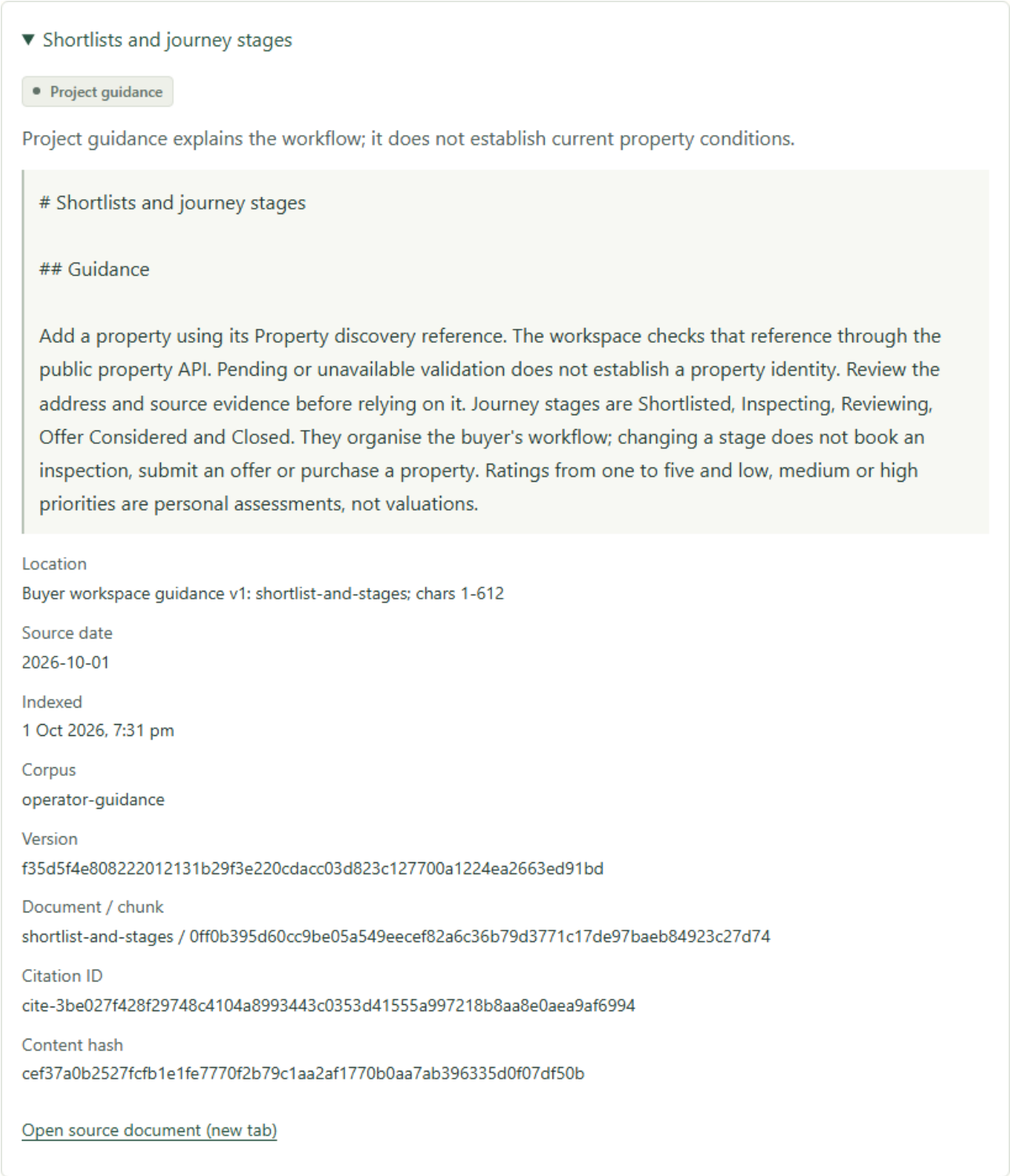


Figure 24 F5 cited passage

PROPERTYSCOPE ASSISTANT

Complete

Insufficient context to provide a grounded explanation.

Confidence: Insufficient

Relevant document context is unavailable or insufficient.

Evidence gaps

Similarity measures retrieval relevance, not factual confidence

A relevant recipe or cooking guidance source specifying the oven temperature, including whether the setting is fan-forced or conventional, is unavailable.

Suggested next step

Sources & activity

Run 4ae48dda... 3 Oct 2026, 12:29 pm [Open full activity](#)

Figure 25 F5 off-topic: cake oven temperature

6.4 Retrieval evaluation

The authored Feature 1 baseline uses a disposable index and the production embedding model.

Measure	Result
Corpus	19 documents, 46 chunks, version da29fbf773df
Embedding	BAAI/bge-small-en-v1.5, 384 dimensions, local CPU
Settings	top-k 5, relevance floor 0.55
Supported questions	27; expected-source recall@5 0.963
Questions the corpus does not cover	4 of 6 retrieve no context
Evaluated	2026-09-26

6.5 GitHub Actions

Each workflow builds its feature's images, runs its tests with coverage and smoke-tests the frontend, API and database in Compose. MCP and RAG stay integrated but disabled; contract tests use doubles:

```
# .github/workflows/student-1.yml ... student-5.yml
env:
  AI_MODE_MCP_ENABLED: "false"
  AI_MODE_RAG_ENABLED: "false"
```

Student 5's workflow also starts host AI-mode offline with an unreachable provider; MCP and RAG remain disabled. All six workflows passed on main at the report's commit reference on 3 October:

Workflow	Successful run
Student 1	37093164388
Student 2	37093164409
Student 3	37093164399

Workflow	Successful run
Student 4	37093164366
Student 5	37093164432
Integration	37093164370

6.6 Docker Compose deployment

The Release 0 `docker-compose.yml` still deploys the whole containerised application. Each backend now also receives the host MCP and RAG addresses beside AI-mode's, for example:

```
f1-backend:
  environment:
    AI_MODE_BASE_URL: http://host.docker.internal:${AI_MODE_PORT:-5005}
    MCP_SERVER_URL: http://host.docker.internal:${MCP_PORT:-5011}/mcp
    RAG_SERVER_URL: http://host.docker.internal:${RAG_PORT:-5012}
```

Backends call AI-mode, which holds the MCP and RAG credentials; containers receive no MCP or RAG token. Compose defines no AI-mode, MCP, RAG or loop service. The [3 October stack status](#) shows all 20 containers running beside the three host processes (extract):

```
Host AI processes (outside Compose)
ai-mode running http://127.0.0.1:5005
mcp running http://127.0.0.1:5011/mcp
rag running http://127.0.0.1:5012
$ docker compose ... --profile release-0 ps
ps-dev-shared-frontend-1 Up (healthy) 127.0.0.1:5100->8080/tcp
ps-dev-f1-frontend-1 Up (healthy) 127.0.0.1:5200->8080/tcp
ps-dev-f2-frontend-1 Up (healthy) 127.0.0.1:5300->8080/tcp
ps-dev-f3-frontend-1 Up (healthy) 127.0.0.1:5600->8080/tcp
ps-dev-f4-frontend-1 Up (healthy) 127.0.0.1:5400->8080/tcp
ps-dev-f5-frontend-1 Up (healthy) 127.0.0.1:5500->8080/tcp
+ 14 backend, database and worker containers: 12 healthy, 2 workers without healthchecks
```

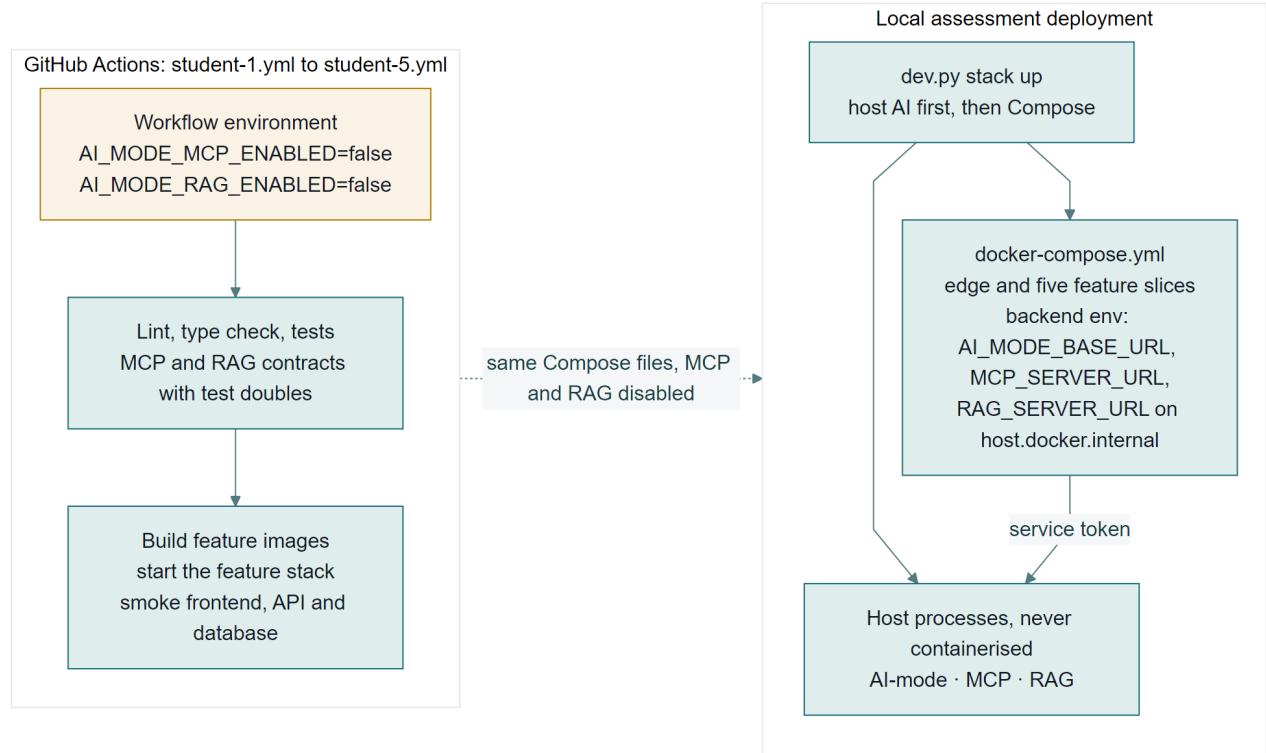


Figure 26 CI and local deployment

7 Integration summary

On 3 October one Compose project and one host AI tier served all five features together. Each reached MCP and its own corpus through its own /api/<feature>/v1/assistant/turns backend route, and its Release 0 page, API and database kept working (6.3).

Feature	Tools returned via MCP	Cited answer	Off-topic question
1	data.releases.v1	Moderate, 2 citations	Insufficient, 0 citations
2	market.cases.inspect.v1, market.sales.summary.v1	Moderate, 2 citations	Insufficient, 0 citations
3	crime.methodology.v1	High, 2 citations	Insufficient, 0 citations
4	duediligence.capabilities.v1, duediligence.review.inspect.v1, duediligence.evidence.summary.v1	High, 3 citations	Insufficient, 0 citations
5	buyer.cases.inspect.v1, buyer.notes.list.v1, buyer.tasks.list.v1, buyer.evidence.collect.v1	Moderate, 2 citations	Insufficient, 0 citations

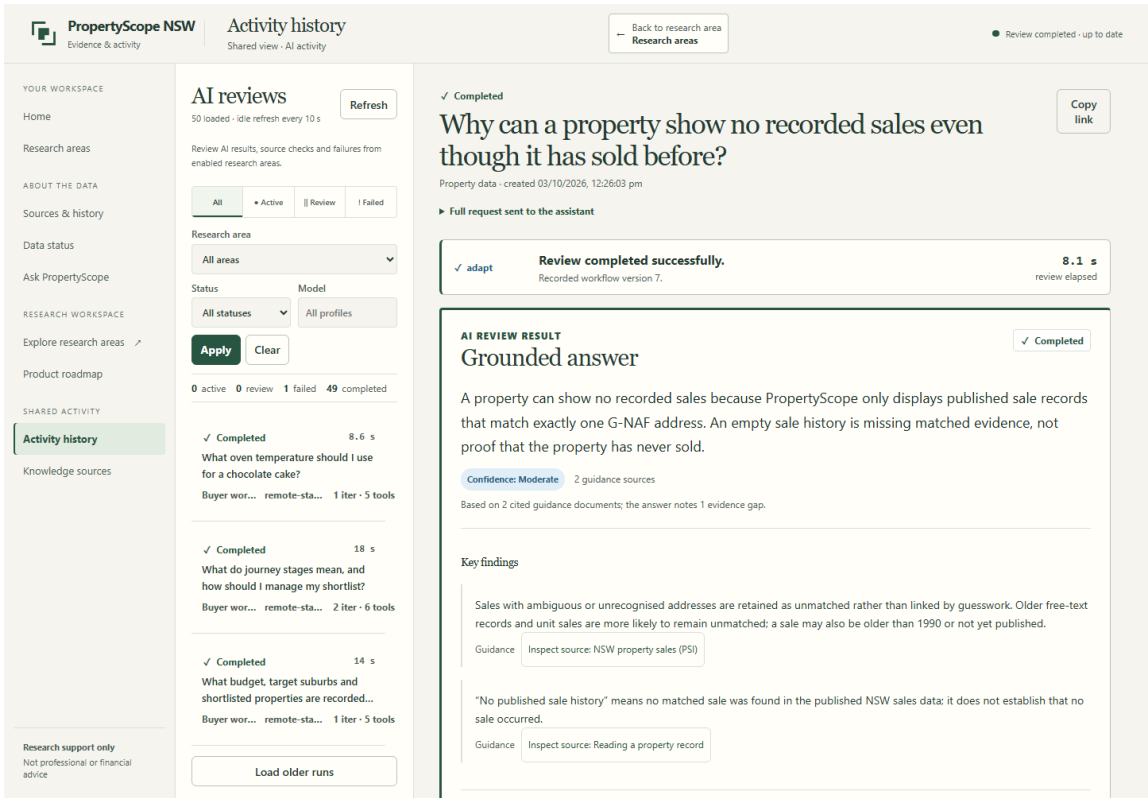


Figure 27 Cross-feature activity history

8 Known issues and limitations

- **Local execution only.** AI-mode, MCP, RAG and the loop are host processes on one machine, bound to loopback; containers reach them through `host.docker.internal`. They are disabled in CI and will not run in the Release 2 cloud deployment.
- **Setup and provider.** First use needs the embedding model and explicit corpus ingestion. Answers need a working OpenAI key; without one, feature CRUD still works.
- **Grounding limits.** Corpora are project guidance, not official methodology; confidence summarises evidence support rather than proving an answer. Irrelevant passages can pass the relevance floor: F2's password question retrieved some, yet correctly returned insufficient context.
- **Data.** The [accepted-data snapshot](#) holds real G-NAF (5,190,134), PSI sales (7,402,643), BOCSAR (318,122) and SEIFA (4,320) records. Schools (10) and fixture properties (3) remain demonstration data. F3 locality coverage is incomplete, and F4's sample review uses synthetic, source-attributed records.
- **Open feature defects.** F2 can keep the previous case's chat history after switching cases. F4's suggested follow-up questions omit some grounded findings, and one exploratory F4 off-topic question failed validation instead of returning insufficient context.
- **Evidence scope.** Captures show selected successful scenarios per feature, not reliability across every question.

9 Contributions, repository and showcase

[Repository](#) and [showcase presentation](#) (public; 9:35 per [player metadata](#)). All five members attended and participated in the 2 October showcase: Matthew Shelton, Burhan Naeem, James Huang, Michael White and Derek Song.

Student 1 Matthew Shelton

Date	Commit	Release 1 work
7 Sep	1e109ac	Shared MCP and RAG servers, host runtime, loop validation modes, F1 grounding
19 Sep	9bc4ee9	Self-service MCP/RAG adoption kit for feature owners
26 Sep	e7485af	Retrieval and confidence calibration, corpus inspection
3 Oct	7fc9b29	Verified MCP transport, readiness fixes, integration evidence, report

Student 2 Burhan Naeem

Date	Commit	Release 1 work
28 Sep	a1b8577	Tool allowlist, guidance corpus, capability tool and shared assistant
28 Sep	d18dfec	Feature evidence; screenshot-capture fix for all features
1 Oct	f019e41	Dedicated assistant workspace; refreshed screenshots

Student 3 James Huang

Date	Commit	Release 1 work
1 Oct	da7556c2	Map displays Feature 1's published data
2 Oct	b1181cd	Suburb and crime tools, shared assistant, guidance corpus
2 Oct	330e65f	Guidance and published locality-context refinements

Student 4 Michael White

Date	Commit	Release 1 work
1 Oct	bbd112d	Grounded tool allowlist and capabilities MCP tool
1 Oct	a6b7b7f	Due-diligence guidance corpus
1 Oct	e0720f4	Shared grounded assistant in site reviews
1 Oct	b4cf74a	Assistant tests in Feature 4 CI
1 Oct	7aa69f6	MCP, RAG and loop evidence

Detailed [owner log](#).

Student 5 Derek Song

Date	Commit	Release 1 work
1 Oct	bb1e3b2	Scoped assistant routes, guidance corpus, cited UI and MCP/RAG evidence

Detailed [owner log](#).